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Received, October 23, 2003.

Accepted, May 24, 2005.

STEREOELECTROENCEPHALOGRAPHY IN THE PRESURGICAL EVALUATION OF FOCAL EPILEPSY: A RETROSPECTIVE ANALYSIS OF 215 PROCEDURES

OBJECTIVE: To report on indications, surgical technique, results, and morbidity of stereoelectroencephalography (SEEG) in the presurgical evaluation of patients with drug-resistant focal epilepsy.

METHODS: Two-hundred fifteen stereotactic implantations of multilead intracerebral electrodes were performed in 211 patients (4 patients were explored twice), who showed variable patterns of localizing incoherence among electrical (interictal/ictal scalp electroencephalography), clinical (ictal semeiology), and anatomic (magnetic resonance imaging [MRI]) investigations. MRI scanning showed a lesion in 134 patients (63%; associated with mesial temporal sclerosis in 7) and no lesion in 77 patients (37%; with mesial temporal sclerosis in 14 patients). A total of 2666 electrodes (mean, 12.4 per patient) were implanted (unilaterally in 175 procedures and bilaterally in 40). For electrode targeting, stereotactic stereoscopic cerebral angiograms were used in all patients, coupled with a coregistered three-dimensional MRI scan in 108 patients.

RESULTS: One hundred eighty-three patients (87%) were scheduled for resective surgery after SEEG recording, and 174 have undergone surgery thus far. Resections sites were temporal in 47 patients (27%), frontal in 55 patients (31.6%), parietal in 14 patients (8%), occipital in one patient (0.6%), rolandic in one patient (0.6%), and multilobar in 56 patients (32.2%). Outcome on seizures (Engel's classification) in 165 patients with a follow-up period of more than 12 months was: Class I, 56.4%; Class II, 15.1%; Class III, 10.9%; and Class IV, 17.6%. Outcome was significantly associated with the results of MRI scanning ($P = 0.0001$) and with completeness of lesion removal ($P = 0.038$). Morbidity related to electrode implantation occurred in 12 procedures (5.6%), with severe permanent deficits from intracerebral hemorrhage in 2 (1%) patients.

CONCLUSION: SEEG is a useful and relatively safe tool in the evaluation of surgical candidates when noninvasive investigations fail to localize the epileptogenic zone. SEEG-based resective surgery may provide excellent results in particularly complex drug-resistant epilepsies.

KEY WORDS: Complications, Epilepsy surgery, Intracerebral electrodes, Outcome, Presurgical evaluation, Stereoelectroencephalography, Stereotaxy

Neurosurgery 57:706-718, 2005

DOI: 10.1227/01.NEU.0000176656.33523.1e

www.neurosurgery-online.com

Refinement of noninvasive techniques, as well as increasing experience in the presurgical evaluation of patients with drug-resistant focal epilepsies amenable to surgical treatment, have significantly decreased the number of cases requiring invasive intracranial electroencephalography (EEG) recording (23, 52). Nevertheless, in a

considerable number of patients, noninvasive investigations fail to localize the site of origin and early spread of the ictal discharge (epileptogenic zone [EZ]). Furthermore, if highly eloquent areas are suspected to be involved in the ictal discharge, functional mapping is mandatory when planning resective surgery close to these regions to avoid new unwanted

postoperative permanent neurological deficits (34, 56). For these reasons, invasive intracranial EEG recording is still indicated in selected patients.

Different techniques of intracranial EEG recording that differ from center to center are currently used, and each has its advantages and drawbacks. Chronically implanted subdural electrodes allow recording from large superficial cortical areas, but they provide limited coverage either of deep-seated structures, such as the hippocampus, or of the cortex within sulci. Intracerebral electrodes have the advantage of an excellent sampling from mesial structures and from the intrasulcal cortex, with the disadvantage of providing information from a limited volume of tissue. Combined use of subdural and intracerebral electrodes also has been advocated (5, 6, 44). Other more or less invasive techniques are represented by epidural electrodes (2) and foramen ovale electrodes (55).

Currently in our center, approximately 45% of resective surgical procedures in patients with drug-resistant focal epilepsy are performed after chronic intracranial EEG monitoring with stereotactically implanted intracerebral electrodes (stereo-electroencephalography [SEEG]), according to the methodology introduced by the group of the S. Anne Hospital, Paris, France. Although several studies on specific issues concerning epilepsy surgery reported series of patients evaluated by SEEG recordings (7, 28, 31, 51), methodological aspects have been addressed only in historical papers that are not easily accessible to readers (1, 48, 49, 50). An exhaustive and updated study on current applications of SEEG, as well as on its role in combination with more recent imaging methods in the investigations of potential candidates to epilepsy surgery, is lacking. The aim of the present retrospective study was to report on indications, technical aspects, results, and morbidity of SEEG in a large consecutive series of patients.

PATIENTS AND METHODS

During the 7-year period from 1996 through 2002, 215 stereotactic procedures of intracerebral electrode implantation were performed at our center in 211 patients with uncontrolled focal epilepsy. Four patients were explored twice. Another eight patients had been investigated previously elsewhere (seven by stereotactically implanted intracerebral electrodes and one by subdural electrodes). There were 120 males and 91 females, whose mean age at seizure onset was 7.8 ± 6.7 years (range, birth–27 yr). At the time of SEEG examination, the mean age was 26.9 ± 11.9 years (range, 2–56 yr), the mean duration of epilepsy was 19.1 ± 10.6 years (range, 1–46 yr), and the mean monthly seizure frequency was 59 ± 109 seizures (range, 1–1000 seizures).

Indications for SEEG Recording

Presurgical noninvasive evaluation, conducted by the clinical neurophysiologists of our center, included: accurate anamnestic investigation mainly focused to type and chronology of ictal symptoms and signs; scalp video-EEG recording of

habitual seizures, with the aim of correlating ictal clinical signs with ictal EEG modifications; brain magnetic resonance imaging (MRI), with particular attention to the region(s) that the electroclinical data indicated as possibly involved in the origin of the ictal discharge; complete neuropsychological examination. We did not routinely perform either interictal positron emission tomography or interictal or ictal single-photon emission computed tomography.

After this stage, a high degree of positive correlation among these investigations regarding lateralization and localization of the EZ, allows the planning of surgery without any further presurgical investigation. Nevertheless in the present group of patients, one of the following situations resulted from the noninvasive workup, thus indicating the need of invasive SEEG recording: 1) with no definite anatomic abnormality at MRI, ictal/interictal scalp EEG findings were partially or fully discordant with ictal clinical semeiology (51 patients); 2) with a well-documented MRI abnormality, ictal/interictal scalp EEG findings and/or ictal clinical semeiology suggested a wide involvement also of extralesional areas (64 patients); 3) whatever the MRI evidences, a) ictal clinical semeiology was discordant with an apparently localizing ictal scalp EEG pattern (16 patients) or b) a quick spread of the discharge to the contralateral hemisphere was observed on ictal scalp EEG (10 patients); 4) either MRI, ictal scalp EEG, or ictal clinical semeiology suggested an early involvement of highly eloquent areas, when their relationships with the EZ had to be established and functional mapping was needed, to define both the prognosis of resection on seizures and the associated surgical risks (40 patients); 5) diffuse, hemispheric, or bilateral MRI abnormalities, with ictal EEG and clinical evidence of more localized/lateralized ictal onset (30 patients). In several patients, SEEG was scheduled for more than one of these indications, the most prominent of them having being reported.

Brain MRI scans disclosed a lesion of variable size in 134 patients (63%), which was associated with suspected mesial temporal sclerosis (MTS) in 7 patients. No definite lesion was detected in 77 patients (37%), in 14 of whom MTS was suggested.

Planning of SEEG Exploration

The available noninvasive anatomic, electrical, and clinical data were reviewed by a clinical neurophysiologist to formulate a coherent hypothesis of localization of the EZ and to plan a consistent tailored strategy of exploration. This was addressed to record from the regions considered the most likely origin of the discharge (including the lesion, if present) as well as from all the structures possibly involved in the organization of the discharge through the more common pathways of propagation, as suggested by available electroclinical findings. Consequently, monobar implantations were very uncommon, and they were limited to two patients with frontal lobe epilepsy. In this way, for instance, patients with suspected frontal seizures with possible insular and rolandic propagation had extensive placement of electrodes in the frontal lobe

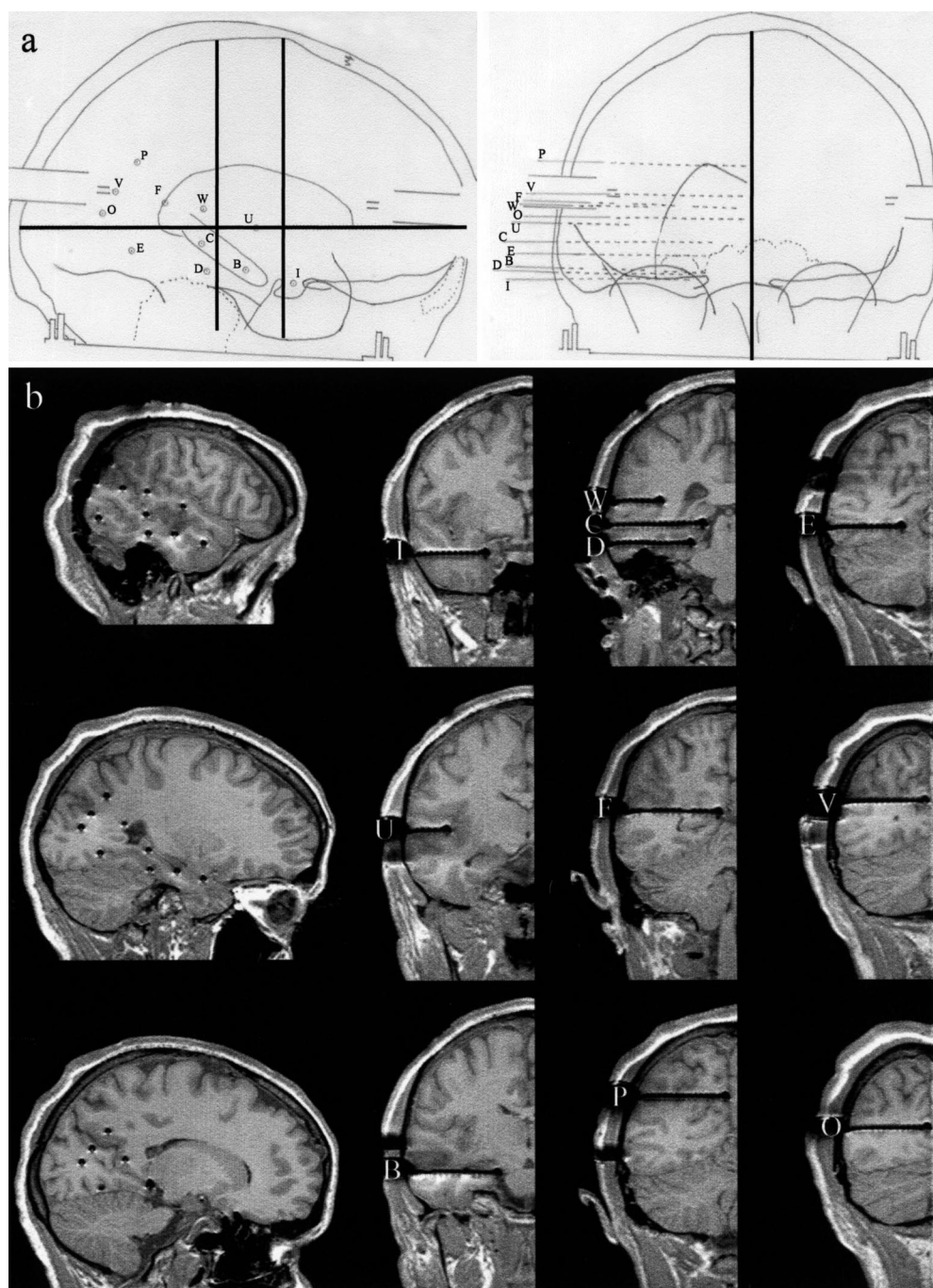


FIGURE 1. A, stereotactic sketch, according to the bicommissural reference system (lateral and anteroposterior views) of a right temporo-occipital SEEG exploration. Electrodes are indicated with either circled dots or dotted lines labeled by uppercase letters. B, T1-weighted three-dimensional postimplantation MRI scan. The electrode arrangement is shown in three sagittal slices (left column), corresponding to superficial (upper box), intermediate (middle box), and deep (lower box) planes. In the nine coronal multiplanar reconstructions (right columns), the electrode location is shown in detail. Uppercase labels correspond to those of the stereotactic sketch.

with additional sampling of the insular and central regions, and patients with seizures possibly originating from the pos-

terior quadrant of the hemisphere had electrodes placed in the posterior temporal, occipital, parietal, and postcentral regions, with different tailored combinations according to individual features of each patient. Although no standard coverages were used, some typical arrangement of electrode placement was used in given areas (Figs. 1 and 2). Rolandic electrodes were implanted when MRI showed anatomic abnormalities within or close to this region and/or when its involvement in the EZ was suspected, with the twofold goal to evaluate the rolandic participation to the ictal discharge and to obtain a functional mapping by intracerebral electrical stimulations.

SEEG Implanting Procedure

A total of 2666 platinum-iridium, semiflexible multilead intracerebral electrodes (diameter, 0.8 mm; 5–18 contacts 1.5 mm in length and 2 mm apart [Dixi Medical, Besançon, France]) were implanted, ranging from 3 to 20 per patient (mean, 12.4 ± 2.7 electrodes/patient). The exploration was unilateral in 175 procedures (right hemisphere in 103 procedures, left hemisphere in 72 procedures) and bilateral (but not symmetrical) in 40 procedures. One or more electrodes were implanted stereotactically in the temporal lobe in 177 procedures, in the rolandic region (precentral and postcentral gyri with the corresponding portion of the operculum and paracentral lobule) in 176 procedures, in the frontal lobe in 154 procedures, in the parietal lobe in 148 procedures, and in the occipital lobe in 92 procedures. The insular cortex was sampled in 34 procedures by electrodes inserted through the suprasylvian or infrasyllian opercula or by a retroinsular trajectory with a frontopolar entry point.

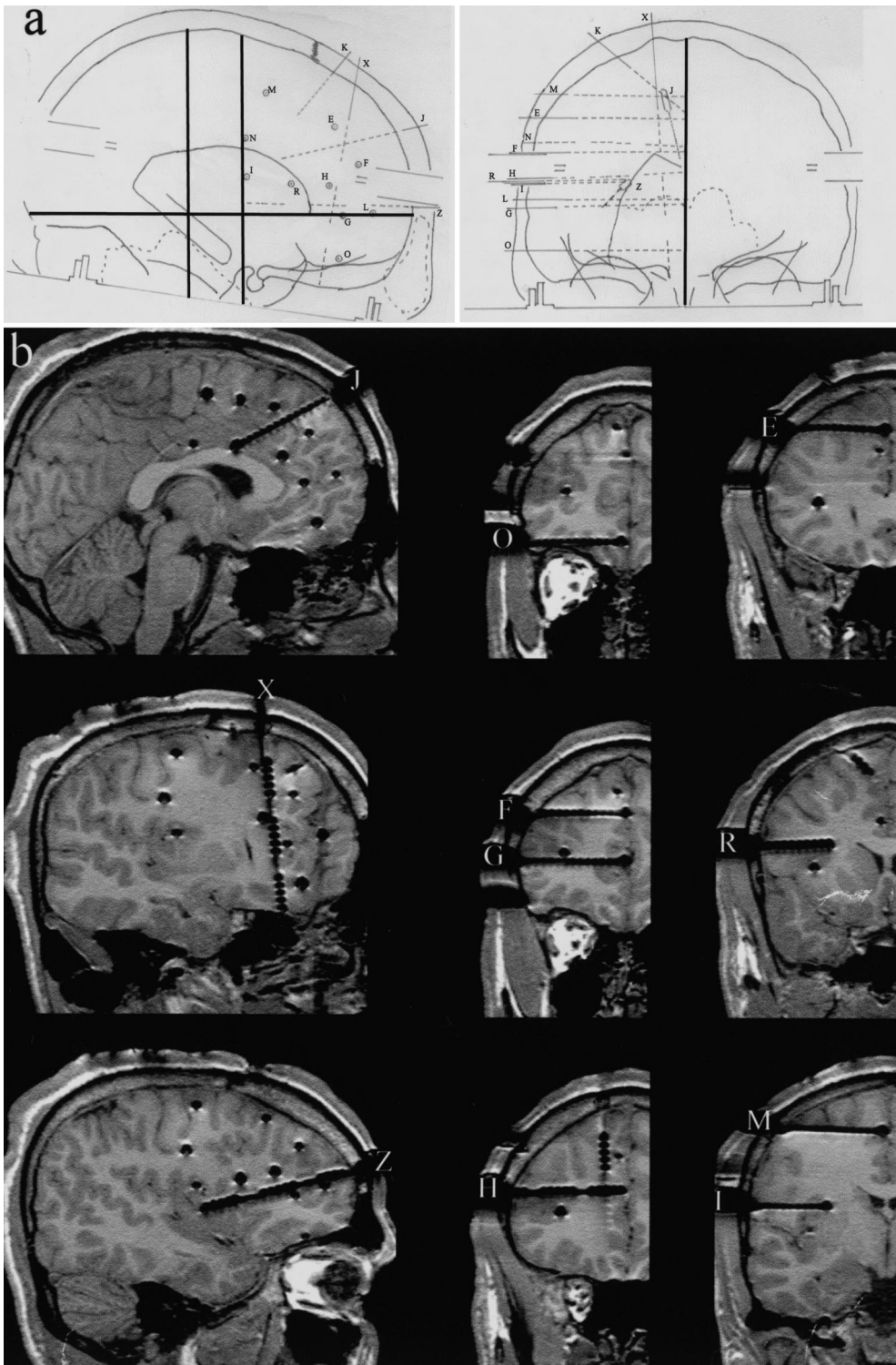


FIGURE 2. A, stereotactic sketch according to the bicommissural reference system (lateral and anteroposterior views) of a right frontal SEEG exploration. Electrodes are indicated with either circled dots or dotted lines labeled by uppercase letters. B, T1-weighted three-dimensional postimplantation MRI scan. Left column contains MRI multiplanar reconstructed slices parallel to the trajectories of electrodes J (upper box), X (middle box), and Z (lower box). In the middle column and right column, coronal multiplanar reconstructions show the location of some of the orthogonally placed electrodes. Uppercase labels correspond to those of the stereotactic sketch.

The stereotactic implantation technique of intracerebral electrodes (SEEG) is described in detail elsewhere (31, 49). In all patients, a cerebral stereotactic stereoscopic teleangiogram (47), by injection of the appropriate vessels through catheterization of the femoral artery, was obtained after placement of the Talairach frame by transosseous pins and under general anesthesia. The stereoscopic angiograms were used in all patients to plan avascular electrode trajectories. In the last 108 patients, the implantation plan was carried out also on three-dimensional MRI (spoiled gradient echo T1-weighted gadolinium-enhanced sequence; slice thickness, 1 mm; no gap; matrix, 512×512) imported in a computer-assisted neuronavigational module (STP 4.0; Leibinger/Fischer, Freiburg, Germany or Voxim; IVS, Chemnitz, Germany), which allowed coregistration with angiographic images. Placement of intracerebral electrodes was performed in a separate procedure under general anesthesia and after refixing the frame exactly in the same position as for the previous stereotactic arteriographic study. A computer-assisted robotized tool holder fully dedicated to stereotactic procedures (NeuroMate; Integrated Surgical Systems, Davis, CA) was used for inserting 42 electrodes in 17 patients. For the implantation of each electrode, a cranial percutaneous trephination was performed with a 2.3-mm twist drill. The dura was perforated by low-current monopolar coagulation. Minor leaks of cerebrospinal fluid often were observed at this stage, but they did not require particular attention. Titanium hollow pegs (diameter, 2.5 mm), for the insertion and the fixation of the electrodes then were screwed to the cranium. A rigid stylet (diameter, 0.8 mm) was advanced through

the peg as far as the established target under fluoroscopic control to trace the intracerebral track of the electrode. After the stylet had been withdrawn, the electrode was inserted and advanced to the target, and it was fixed to the peg by a plastic cap. The cables were sutured to the skin and a sterile medication applied. Antibiotics were not administered until June 2001 (157 procedures), when their use was started according to our hospital's guidelines, consisting of a single bolus of cephamezine (1–2 g intravenously, depending on patient's weight) at anesthesia induction (58 procedures). An MRI scan with implanted electrodes was obtained in all patients to define the actual position of each electrode contact.

Video SEEG Recording

After implantation, video SEEG monitoring began, with the purpose of recording patient's habitual ictal manifestations. After an adequate number of seizures was obtained, the patients underwent the sessions of intracerebral electrical stimulations. High- and low-frequency electrical stimulations were delivered to pairs of contiguous contacts, with the twofold purpose of inducing habitual ictal manifestations and providing a functional map of the implanted regions (e.g., motor, sensitive, visual, language areas) (22, 32, 48). The mean duration of video SEEG monitoring was 10.7 ± 4.5 days (range, 2–29 d). After monitoring was completed, electrodes were withdrawn under local anesthesia or mild sedation. Electroclinical data were reviewed separately by two neurophysiologists, and this process resulted in the definition of the EZ as the area characterized by the presence of a low-voltage fast-activity discharge before or concurrent with ictal clinical symptoms.

Evaluation of the Surgical Results

For patients who subsequently underwent surgery for resection of their EZ and who were followed up for at least 1 year, outcome on seizures was assessed according to the Engel's classification (12). Univariate analysis was performed comparing seizure outcome (Engel's Class I versus Classes II–IV) with discrete numerical variables (age at seizure onset, illness duration before surgery, monthly seizure frequency) by the *t* test. Univariate χ^2 analysis was performed to compare seizure outcome (Class I versus Classes II–IV) against the categorical variables (sex, presence of lesion on MRI, presence of MTS on MRI, side of SEEG, site of EZ, site of resection, type of resection, completeness of lesionectomy, reason for incomplete lesionectomy, histological diagnosis, presence of MTS at histological analysis on resected mesial temporal specimens). A stepwise multivariate logistic regression analysis was performed by using all the considered numerical and categorical variables.

RESULTS

Targeting of desired structures, as demonstrated by postimplantation MRI scans, was generally satisfactory using stereo-

tactic stereoscopic angiography. Nevertheless, the use of coregistered volumetric MRI images (108 procedures) improved the precision of targeting deep structures and lesions (Fig. 3).

A minimum of two contacts per electrode were recorded from cortical structures, but a wide range of variability occurred from this point of view, considering that electrodes often encroached on the cortex tangentially to the slope of the gyral surface, that cortical thickness is inhomogeneous, and that in many instances, external, intermediate, and internal cortical structures were explored (Fig. 4). Slight displacement of electrodes during recording was an uncommon occurrence that was detected easily by significant changes of electrical activity in the EEG trace of bipolar derivations between contiguous contacts. This was simply corrected by replacing the electrode to its original depth under fluoroscopic control.

SEEG Morbidity

No deaths related to the SEEG procedure and no morbidity associated to angiographies occurred in this series. Complications were observed in 12 of 215 procedures (5.6%). They are summarized in Table 1.

Nine (4.2%) patients experienced hemorrhagic complications. Three clinically evident intracerebral hematomas that required emergency evacuation of the clot left the patients with mild (one patient) to severe (two patients) neurological impairment. In five patients, asymptomatic intracranial bleedings (two intracerebral, three subdural) of limited size were detected incidentally on postimplantation MRI or computed tomographic imaging scans. The remaining patient (a young woman with valproate-related platelet aggregation abnormalities) experienced acute hydrocephalus for an intraaqueductal clot that was treated successfully with temporary external cerebrospinal fluid drainage.

A possible infective complication occurred in one (0.5%) patient. This patient, who did not receive antibiotic prophylaxis, experienced progressive motor impairment of her left arm 4 weeks after electrode removal. Computed tomographic

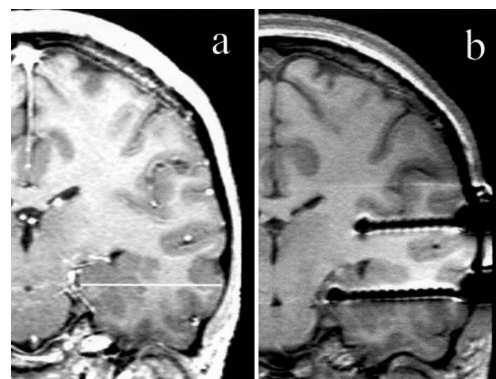


FIGURE 3. A, gadolinium-enhanced, T1-weighted coronal slice MRI scan showing planning of an electrode trajectory intended to sample a temporal periventricular heterotopy and the adjacent hippocampus. B, T1-weighted coronal slice MRI scan at the same level obtained after electrode placement showing successful targeting of both structures.

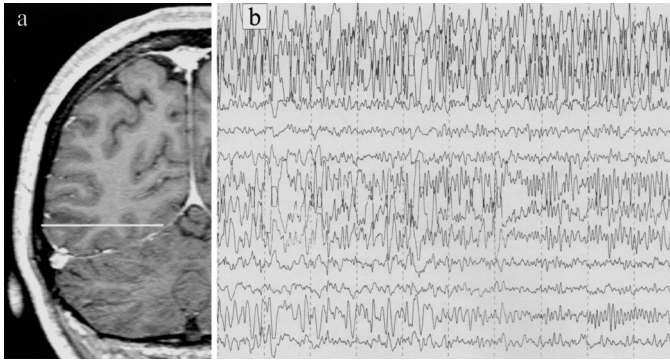


FIGURE 4. A, gadolinium-enhanced, T1-weighted coronal slice MRI scan showing planning of an electrode trajectory intended to sample from temporal lateral, basal and mesial cortex (electrode E in Fig. 1B). B, intracerebral SEEG traces recorded from the electrode contacts show distinctive features of cortical activity in most derivations.

imaging and MRI scans showed a picture suggesting a cerebral abscess in the right rolandic region. No culture was obtained. Full-dose antibiotic treatment resulted in disappear-

ance of the lesion at neuroradiological follow-up 4 months later, with complete neurological recovery.

Other causes occurring in two patients accounted for a further 1% rate of complications. A transient sensory deficit in one (0.5%) patient's left hand was ascribed to a small area of cortical edema at the entry point of an electrode, revealed on MRI to be in his contralateral right parietal region. In another patient (0.5%), intracranial rupture of an electrode implanted in the right temporal pole required its surgical removal.

Localization of the EZ

SEEG monitoring resulted in poor definition of the EZ in five patients, who declined further invasive investigations. In another two patients, SEEG evaluation was not completed because of major complications detected during or immediately after electrode implantation that required electrode removal and emergency evacuation of an intracerebral hematoma. In the remaining 204 patients, intracerebral recording allowed the definition of the EZ, whose localizations are reported in Table 2. Two of these patients required two explorations for a correct definition of the EZ, and two patients

TABLE 1. Clinical details of the 12 patients with a stereoelectroencephalography complications^a

Patient no.	Sex/age (yr)	Side of SEEG	Lobes of SEEG	No. of electrodes	Complication	Symptoms	Treatment	Resulting deficit
1 ^b	F/37	R	FrTCP	13	R: rolandic IC hematoma	L hemiplegia	Craniotomy, evacuation of clot	L: hemiparesis
2 ^b	F/21	R	FrTCP	13	R: rolandic IC hematoma	No intracranial EEG background activity	Craniotomy, evacuation of clot	L: hemiparesis
3	M/34	R	TPC	16	R: parietal IC hematoma	Bleeding from screw during procedure	Craniotomy, evacuation of clot	Mild astereognosia
4 ^c	M/46	L	FrTP	11	L: frontal IC clot	No	No	No
5	M/22	R	FrT	12	R: frontocentral SD clot	No	No	No
6	M/28	R	FrTCP	16	R: central parietal SD clot	No	No	No
7	M/13	Bilateral	T (R)T (L)	16	L: temporopolar IC clot	No	No	No
8	M/41	R	FrTCPO	16	R: parietal SD clot	No	No	No
9	F/27	R	TOP	14	Clot in aqueduct, acute obstructive hydrocephalus	Progressive impaired consciousness	External CSF drainage	No
10	F/45	Bilateral	FrC (R)Fr (L)	12	R: rolandic cerebral abscess	Progressive L motor deficit	Antibiotics	No
11	M/27	R	FrC	10	R: rolandic focal cortical edema	Sensory deficit in L hand	No	No
12	M/13	R	FrTC	13	Retained broken electrode in T pole	No	Surgical removal of electrode	No

^a SEEG, stereoelectroencephalography; R, right; Fr, frontal; T, temporal; C, central; P, parietal; IC, intracerebral; L, left; EEG, encephalography; SD, subdural; O, occipital; CSF, cerebrospinal fluid. ^b SEEG not completed, no surgery.

^c Poorly localizing SEEG, no surgery.

TABLE 2. Localization of the epileptogenic zone as revealed by stereoelectroencephalography recording of 204 patients)^a

Site of epileptogenic zone	No. of patients
Temporal	51
<i>M + L</i>	45
<i>L</i>	6
Frontal	56
<i>Olofrontal</i>	9
<i>Polar-orbital</i>	6
<i>M + L</i>	9
<i>M</i>	26
<i>L</i>	6
Parietal	15
<i>M + L</i>	6
<i>M</i>	6
<i>L</i>	3
Occipital	1
Central	5
Multilobar	62
Multifocal/bilateral	12
Hemispheric	2

^a M, mesial; L, lateral.

underwent repeated SEEG after failed resective surgery. Nineteen of 204 patients were excluded from surgery for electroclinical reasons, which are summarized in *Table 3*. One hundred eighty-five patients were scheduled for surgery. In two of them a stabilized motor deficit was present, the ictal discharge simultaneously started over a wide hemispheric area and the patients were referred to other centers for hemispherectomy. Resective surgery has been performed in 174 patients, six patients are awaiting surgery, two patients declined further treatment, and one patient died from causes unrelated to seizures while waiting for surgery.

Functional Mapping

Intracerebral electrical stimulations performed for functional purposes in 209 patients who completed SEEG recording allowed the identification of regions related to the following critical functions in 171 patients: primary somatomotor (138 patients), primary somatosensory (32 patients), visual (66 patients), and language (19 patients). Multiple-function mapping was obtained in 74 patients. In most patients, positive responses were obtained from stimulations both in gray and

TABLE 3. Exclusion from surgery (19 patients)

Stereoelectroencephalography evidence	No. of patients
Bilateral epileptogenic zone	8
Multifocal epileptogenic zone	4
Epileptogenic zone involving highly functional areas	7

white matter, thus allowing the mapping of critical pathways extensively. No responses were obtained in 38 patients who had electrodes placed far from eloquent areas or who did not provide adequate cooperation.

Resective Surgery

The sites and type of surgical resections in the 174 operated patients are reported in *Tables 4* and *5*, respectively. The reasons for 53 incomplete lesion removals are summarized in *Table 6*. Findings of histological evaluations on resected specimens, according to main pathological categories, are shown in *Table 7*. All 19 patients with MRI evidence positive for MTS had their hippocampus removed and, in 14 of them, the diagnosis was histologically confirmed. Histological results were also positive for MTS in 13 patients with no hippocampal abnormalities on MRI scans. Among the 46 patients who underwent surgery and who had negative MRI results (no lesion, no MTS), only in 6 patients was histological examination was unrevealing.

Results of Resective Surgery on Seizures

One-hundred-sixty-five patients have a postoperative follow-up period of at least 1 year, and the results of surgery on seizures are summarized in *Table 8*.

Preoperative MRI features showed a statistically significant association with outcome both at univariate ($P = 0.001$) and at multivariate analysis, as well as completeness of lesionectomy ($P = 0.038$). The likelihood ratio test value for the logistic regression model was 25.27 ($P = 0.00001$). For the 50 patients who received incomplete lesionectomy, univariate analysis showed a statistically significant association between seizure outcome and the reasons of resection incompleteness ($P = 0.0002$).

DISCUSSION

Recently, there has been renewed interest in the surgical treatment of drug-resistant focal epilepsies, with an increasing number of potential candidates being referred to dedicated centers for presurgical investigation. The growing experience of epilepsy surgery teams in evaluating electroclinical results,

TABLE 4. Lobes of surgical excision in 174 patients who underwent surgery

Tailored resections	No. of patients	% of patients
Unilobar	118	67.8
<i>Temporal</i>	47	27.0
<i>Frontal</i>	55	31.6
<i>Parietal</i>	14	8.0
<i>Occipital</i>	1	0.6
<i>Rolandic</i>	1	0.6
Multilobar	56	32.2

TABLE 5. Type of resection in 174 patients who underwent surgery

Type of resection	No. of patients	% of patients
Corticectomy	58	34
Corticectomy + lesionectomy	112	64
Complete ^a	62	
Incomplete ^a	51	
Lesionectomy	3	2
Complete	1	
Incomplete	2	

^a Referred to as *lesion removal*.**TABLE 6. Reasons for incomplete lesion removal in 53 patients**

Incomplete lesion removal	No. of patients
Deliberate	45
<i>Lesion partially involved in epileptogenic zone^a</i>	16
<i>Lesion involving eloquent areas</i>	18
<i>Large hemispheric/multiple lesions</i>	8
<i>Planned two-step surgery^b</i>	3
Unwanted (insufficient resection)	8

^a For nontumoral lesions.^b Only first step performed so far.**TABLE 7. Histological results, according to main pathological categories, in 174 patients who underwent resective surgery**

Histological results	No. of patients ^a	% of patients
Nonneoplastic		
<i>Focal cortical dysplasia</i>	76 (11)	43.7
<i>Other malformative</i>	32 (10)	18.4
<i>Degenerative</i>	16 (1)	9.2
<i>Cryptogenic</i>	18 (4)	10.3
Neoplastic		
<i>Glioneuronal tumors</i>	22	12.7
<i>Other tumors</i>	10 (1)	5.7

^a Numbers in brackets indicate presence of MTS on mesiotemporal specimens.

along with the development of high-resolution MRI, makes surgery available to a large number of patients on the basis of noninvasive presurgical investigations alone (13, 52). This is because of the occurrence of a high degree of concordance among all electrical (interictal/ictal surface EEG), clinical (ictal semeiology), and anatomic (MRI) features of each patient regarding localization of the EZ.

Nevertheless, a considerable number of patients who could benefit from surgery do not show clear-cut evidence of localization of the ictal onset after noninvasive evaluations. In these patients, invasive studies with intracranial EEG monitoring is indicated. This is the case for many patients with no definite lesions on MRI, with the exception of those with MRI results suggesting MTS associated with unequivocal EEG and clinical ictal features of temporal lobe epilepsy. However, the degree of concordance or discordance among anatomic, electrical, and clinical findings that indicate a need for invasive recordings is different among centers (56). Although demonstration of definite abnormalities on MRI often provides helpful information for planning surgery based only on noninvasive diagnostic studies, it is our opinion that several patients with MRI results suggesting a lesion and with poor localization or inconsistent electroclinical findings need to be investigated by invasive tools for an optimal definition of the EZ. It must be stressed, in fact, that the EZ should not be considered as a mere anatomic entity corresponding to a structural abnormality detected by MRI, but it may be better defined in functional terms as the result of the complex and still poorly understood interactions between an underlying pathological process and the epileptogenic network involved in the generation and spread of the ictal discharge. That is, the relationships between the lesion and the EZ need to be assessed accurately to maximize the potential benefit of resective surgery, because the region of onset and early propagation of the discharge generating a seizure may extend well beyond the limits of the lesion (30). However, negative findings on preoperative MRI do not per se justify exclusion from surgical consideration (8), especially in patients with electroclinical features that help to predict an ictal onset localization (42) and in whom histological results often are positive, as in several patients presented herein. These assumptions are probably the best explanation for the fairly high percentage of patients (approximately 45%) undergoing invasive recording with stereotactically implanted intracerebral electrodes (SEEG) in our center compared with other studies (6), although a comparable (3, 4, 39) or even higher (46) rate of invasive tests has been reported.

As far as functional imaging is concerned, we did not include routine interictal positron emission tomography or ictal/interictal single-photon emission computed tomography in the presurgical evaluation because these tests were not available in our hospital during the study period. Nevertheless, our group previously reported poor topographical correlations between metabolic abnormalities, as revealed by interictal fluorodeoxyglucose-positron emission tomography, and both ictal and interictal electrical patterns detected by SEEG recording, especially in extratemporal epilepsies (27).

SEEG methodology implies a rigorous preimplantation scrutiny of all available findings obtained during the noninvasive phase to define a coherent hypothesis as to the likelihood of localization of the EZ. In this decision-making process, the respective weight of MRI, interictal/ictal EEG, and ictal clinical characteristics may vary greatly depending on the patient. After a localizing hypothesis is formulated, a tailored

TABLE 8. Outcome in 165 patients with postoperative follow-up of more than 12 months^a

Variables	No. of patients	No. of cases in Engel's class (%)				
		SF ^b	I	II	III	IV
All patients	165	73 (44.2)	93 (56.4)	25 (15.1)	18 (10.9)	29 (17.6)
Preoperative MRI ^c						
Lesion	104	50 (48.1)	59 (56.7)	17 (16.3)	13 (12.5)	15 (14.5)
Lesion + MTS	7	5 (71.4)	7 (100)	—	—	—
MTS alone	12	9 (75.0)	11 (91.7)	1 (8.3)	—	—
Negative	42	12 (28.6)	16 (38.1)	7 (16.7)	5 (11.9)	14 (33.3)
Site of resection ^d						
Temporal	44	24 (54.5)	32 (72.7)	5 (11.4)	2 (4.5)	5 (11.4)
M + L	41	22 (53.7)	30 (73.2)	5 (12.2)	2 (4.9)	4 (9.7)
L	3	2 (66.7)	2 (66.7)	—	—	1 (33.3)
Frontal	52	24 (46.2)	29 (55.8)	7 (13.4)	8 (15.4)	8 (15.4)
Olofrontal	9	4 (44.4)	5 (55.6)	—	3 (33.3)	1 (11.1)
Polar/orbital	5	3 (60.0)	3 (60.0)	—	1 (20.0)	1 (20.0)
M + L	7	4 (57.1)	4 (57.1)	1 (14.3)	1 (14.3)	1 (14.3)
M	25	10 (40.0)	14 (56.0)	6 (24.0)	2 (8.0)	3 (12.0)
L	6	3 (50.0)	3 (50.0)	—	1 (16.7)	2 (33.3)
Parietal	14	7 (50.0)	7 (50.0)	1 (7.1)	—	6 (42.9)
M + L	5	2 (40.0)	2 (40.0)	—	—	3 (60.0)
M	6	2 (33.3)	2 (33.3)	1 (16.7)	—	3 (50.0)
L	3	3 (100)	3 (100)	—	—	—
Other unilobar	2	—	—	1 (50.0)	—	1 (50.0)
Multilobar	53	18 (34.0)	25 (47.2)	11 (20.7)	8 (15.1)	9 (17.0)
Type of resection ^e						
Corticectomy	54	21 (38.9)	27 (50.0)	8 (14.8)	5 (9.3)	14 (25.9)
Corticectomy + complete lesionectomy	60	37 (61.7)	42 (70.0)	8 (13.3)	8 (13.3)	2 (3.4)
Corticectomy + incomplete lesionectomy	48	14 (29.2)	23 (47.9)	9 (18.8)	5 (10.4)	11 (22.9)
Complete lesionectomy	1	1 (100)	1 (100)	—	—	—
Incomplete lesionectomy	2	—	—	—	—	2 (100)
Reason of incomplete lesionectomy ^f	50					
Partially involved in epileptogenic zone	15	7 (46.7)	13 (86.7)	2 (13.3)	—	—
Involving eloquent areas	16	4 (25.0)	7 (43.8)	4 (25.0)	—	5 (31.2)
Diffuse/large	8	1 (12.5)	1 (12.5)	1 (12.5)	2 (25.0)	4 (50.0)
Two-step surgery	3	—	—	—	2 (66.7)	1 (33.3)
Insufficient resection	8	2 (25.0)	2 (25.0)	2 (25.0)	1 (12.5)	3 (37.5)
Histological results ^d						
Focal cortical dysplasia	72	36 (50.0)	41 (57.0)	5 (6.9)	8 (11.1)	18 (25.0)
Other malformative	31	14 (45.2)	18 (58.1)	7 (22.6)	4 (12.9)	2 (6.4)
Degenerative	16	2 (12.5)	8 (50.0)	3 (18.8)	3 (18.8)	2 (12.4)
Cryptogenic	16	8 (50.0)	9 (56.3)	4 (25.0)	—	3 (18.7)
Glioneuronal tumors	21	11 (52.4)	13 (61.9)	3 (14.3)	1 (4.8)	4 (19.0)
Other tumors	9	2 (22.2)	4 (44.4)	3 (33.4)	2 (22.2)	—
Mesial temporal sclerosis shown in histological analysis ^d	49 ^g					
Present	27	14 (51.9)	18 (66.7)	5 (18.5)	2 (7.4)	2 (7.4)
Absent	22	9 (40.9)	13 (59.1)	4 (18.2)	2 (9.1)	3 (13.6)

^a SF, seizure free; MRI, magnetic resonance imaging; MTS, mesial temporal sclerosis; M, mesial; L, lateral.^b Includes only patients in classes Ia and Ic (no auras).^c $P = 0.001$.^d Statistically not significant differences ($P > 0.05$).^e $P = 0.038$.^f $P = 0.0002$.^g Hippocampectomy was performed in 56 (either temporal or multilobar resections) of these 165 patients, but 7 specimens were inadequate for mesial temporal sclerosis assessment.

implantation strategy is planned, with the goal of confirming or rejecting the hypothesis. In this view, the exploration is focused to sample the anatomic lesion (if present), the more likely structure(s) of ictal onset, and the possible pathway(s) of propagation of the discharge. This accounts for the apparent overlapping of explored lobes among our patients, who received intensive sampling of the region of predicted ictal origin, with a variable number of electrodes placed in contiguous or functionally related structures to evaluate the spread of the discharge adequately. The desired targets then were reached with the precision of the stereotactic technique, allowing them to be recorded from either lateral, intermediate, or deep structures in a three-dimensional arrangement, thus accounting for the dynamic, multidirectional spatiotemporal organization of the ictal discharge. On these grounds, SEEG differs substantially from depth recording, which relies on bilateral and symmetrical placement of intracerebral electrodes in standardized anatomic targets, with the assumption that one should "avoid biasing the exploration strategy in favor of one's preferred localizing hypothesis" (16). With these methodological premises, SEEG, although offering the advantages of a stereotactic three-dimensional concept of the ictal discharge, has the disadvantage of tunnel vision (36), as opposed to subdural arrays, which allow following the development of the epileptic discharge over the surface of a large portion of cerebral cortex. As suggested by others, the judicious combination of stereotactically implanted intracerebral and subdural electrodes may be a helpful option in selected patients (5, 6, 44).

In 7 of our patients (3.3%), SEEG failed to define the EZ, indicating that the previously elaborated hypothesis of localization was incorrect. The reported rate of poor localization after invasive recordings amounts to 10 to 20% of the studied patients (4, 42). In this regard, no consensus exists regarding whether such patients should be considered for repeated invasive tests (43). The two patients in our series who received a double exploration before surgery had, at their first evaluation, ictal SEEG evidence of an EZ partially covered by the exploration at the margins of the electrode arrangement, which was corrected in the second step. The other poorly localized patients declined further investigations. In our patient population, SEEG recording contributed to the localization of the EZ in 204 of 211 patients.

One may argue whether SEEG is worthwhile for indicating hemispherectomy, as in two of our patients. In one of them, who subsequently experienced the clinical and MRI features of Rasmussen's encephalitis, we would not perform SEEG anymore. The other patient, a young man with hemispheric postischemic atrophy and stable contralateral motor deficit, did not differ substantially from other patients in this series with diffuse, hemispheric, or even bilateral malformative or degenerative abnormalities in whom electroclinical findings were consistent with a more localized ictal onset and who could benefit from limited resections after SEEG monitoring. In the other 202 patients, SEEG allowed localization of the onset of the ictal discharge. Patients with multifocal and/or

bilateral foci were excluded from surgery, according to the universally agreed assumption that epilepsy surgery is indicated in patients with a single and stable EZ. It is conceivable that stricter selection criteria should be adopted to rule out invasive investigations in patients with suspected multiple ictal onsets (42).

Chronic intracerebral seizure monitoring coupled with functional mapping was crucial in distinguishing between patients with early ictal involvement of highly eloquent regions, who were excluded from surgery (seven patients), and those with later spread of the discharge to these structures who could undergo surgery with limited surgical risks and with predictable benefit on seizures. Functional mapping allows anticipation of potentially acceptable postoperative deficits, such as visual field defects in posterior temporal, occipital, and parietal resections, as well as evaluation of the risk-to-benefit ratio of excisions close to more critical regions, such as sensory motor and language areas. Although similar functional information can be obtained from acute intraoperative electrical cortical stimulation also, the following points must be stressed. First, chronic SEEG intracerebral electrical stimulations makes mapping of both cortex and fibers feasible, allowing the planning of safer resections also in critical subcortical areas. Second, availability of functional information before and not during surgery allows the patient to participate in the discussion of the risk-to-benefit balance in a relaxing and comfortable setting. Third, intraoperative electrocorticography only exceptionally results in spontaneous seizure recording, which is essential in evaluating the actual ictal involvement of eloquent areas.

One hundred eighty-three patients were scheduled for resective surgery, and 174 of them have undergone surgery so far. Temporal lobe resections were performed in approximately 25% of the studied patients. In several of these patients, the indication to SEEG recording was formulated at the very beginning of the activity of this center and, as the experience of the team has increased with time, the rate of temporal lobe resections performed after SEEG has dropped to 16%, with invasive tests having been reserved for patients with atypical ictal clinical features (33) or with poorly localizing ictal EEG evidence. In most patients (75%), SEEG indicated an extratemporal or multilobar ictal onset, a proportion that is comparable with other studies (25, 35), although several reports on invasive EEG tests show a preponderance of temporal lobe epilepsies over other localizations (6, 20, 37, 54).

Overall, surgical results on seizures show that 56% of our patients were in Class I, with 44% of them being completely seizure free. In our noninvasively evaluated group of patients operated on in the same period of the present study, 83% are in Class I and 69% of patients are completely seizure free (data not shown). It has been reported that the use of invasive EEG predicts a nonfavorable outcome for seizures compared with surgery performed after a noninvasive workup (21), a finding that probably reflects the particular complexity of patients who need intracranial monitoring. In our series, approximately two thirds of patients had frontal or multilobar epilep-

sies, which have a reputation of less favorable results after surgery (26, 40), accounting for the specific challenges associated with these conditions, although our data do not show statistically significant differences of outcome comparing different sites of ictal onset.

Outcome was correlated with preoperative MRI findings, with significantly better results having been found in patients with positive MRI either for a lesion or for MTS, as reported by others (15, 21, 29). Nevertheless, a substantial proportion of patients with normal MRI results (nearly 40% in our series) may benefit from resective surgery (8, 42), indicating that these patients should not be excluded a priori from further invasive presurgical investigations, as advocated by others (41).

A significant relationship was found between complete lesionectomy and Class I outcome. Although there seems to be wide agreement on this point (10, 24), controversial results have been reported (9, 19), raising the possibility that different roles in epileptogenesis are played by different types of lesions and that a structural abnormality may be incompletely involved in the generation of ictal activity. It is beyond the goals of the present study to address this issue, but our results indicate that SEEG is helpful in distinguishing between epileptogenic and nonepileptogenic areas within the same lesion, thus allowing for the control of seizures even with incomplete lesion removals in selected patients. Nevertheless, a substantial number of patients remain in whom incomplete lesionectomy either because of insufficient surgery, functional limits, or extension of the structural abnormality to large hemispheric areas prevents obtaining favorable results.

When facing a mere diagnostic procedure, one should be aware of the potential risks related to the invasiveness of the technique used. Overall SEEG morbidity was 5.6% in this series, a figure comparable with the results reported in most published series dealing with morbidity of intracranial EEG recording, which ranges from 0 to 26%. Most of the complications in our patients were hemorrhagic (4.2%), which has been reported in several studies to be the most frequent adverse event with intracerebral electrodes (11, 13, 17, 50, 53), although intracranial bleeding also has been associated with subdural electrodes (18, 25, 39). Nevertheless, only three of our patients with intracranial bleeding required emergency evacuation of the clot, with resulting severe motor impairment in two of them (1%). A case of posthemorrhagic obstructive hydrocephalus was observed in a patient with a preexisting valproate-related mild platelet aggregation abnormality. Since the observation of this patient, we have paid particular attention in the presurgical screening of hemostasis parameters in patients who are candidates for electrode implantation and have withdrawn administration of valproic acid in the weeks preceding surgery. A possible brain abscess has been detected in one patient who did not receive antibiotic prophylaxis. No further infections were recorded after introduction of prophylactic antibiotic therapy, although its role in preventing infections after placement of intracranial electrodes is at least contradictory (13, 14, 17, 38, 45, 53). Displacement of electrodes has been reported during agitated seizures either with sub-

dural (37, 39) or intracerebral (5, 17) invasive monitoring. In our experience, abrupt changes of electrical activity recorded from the same electrode occasionally may be ascribed to its displacement which, if confirmed by MRI, can be corrected easily with electrode replacement. In one of our patients, displacement of an electrode resulted in its rupture under the cranial vault, and the retained fragment was removed surgically with no consequences. Finally, brain edema has been reported to complicate subdural grid placement (18, 25). The rather uncommon occurrence of focal cortical edema in one of our patients can be ascribed to either local injury, discrete infectious inflammatory reaction, or local ischemic damage.

In conclusion, we believe that the presented approach accounts for the favorable results regarding seizure control observed in our patients operated on after SEEG evaluation, with 44% of patients being completely seizure free and more than 80% of patients with significant improvement (Engel's Classes I–III). In connection with the potential benefits of SEEG recording, a risk as high as 1% of major complications resulting in severe permanent neurological deficits must be taken into consideration. SEEG represents a useful and relatively safe tool that allows for further treatment and potential benefits to a large number of otherwise intractable epileptic patients.

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Acknowledgments

The authors did not receive any financial support from the manufacturers of medications, materials, and devices described in this article. The results of the present study were presented in part at the 12th European Congress of Neurosurgery, Lisbon, Portugal, September 2003.

COMMENTS

In this interesting manuscript, Dr. Cossu and colleagues present a retrospective study of a large series of patients who underwent stereo-electroencephalography evaluation (SEEG). SEEG provided additional indication towards epileptic focus resection in 183 patients (87%), resulting in seizure-free outcome in 44% of the cases, and an overall significant improvement in 82%. Major complications occurred in less than 1% of the patients, with an overall hemorrhagic event risk of 4.2%. The authors concluded that SEEG is a useful and relatively safe tool in the presurgical evaluation of focal epilepsy.

In a small subgroup of patients with intractable focal epilepsy, invasive physiological evaluation may be required to better define the epileptogenic zone. Subdural grids, SEEG, depths electrodes, or a combination of these methods currently are available. The method of choice depends on the presumed location and extension of the epileptogenic zone defined by noninvasive tests. The complication rate always will be a significant factor in the choice of any given invasive methodology.

In the United States, no major epilepsy center is using SEEG. We feel that the extremely low complication rate, coupled with a relatively high success rate reported in this manuscripts, should encourage some of the major epilepsy centers in the United States to apply this technology in selected cases with intractable focal epilepsy. This applies particularly to those cases in which the noninvasive evaluation suggests a deep-seated epileptogenic zone that cannot be assessed accurately with subdural electrodes.

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Accurate and safe lateralization and localization of the ictal onset zone is a high priority in epilepsy surgery. The authors describe their extensive experience with the use of stereotactic implantation of depth electrodes for SEEG recording. This highly specialized technique is used safely and effectively by the authors, with a morbidity rate of 5.6% and 1% permanent severe deficits.

The SEEG technique is a useful one, with advantages over subdural electrode recording for deeper targets such as the insula, deep areas of dysplasia, and along deep sulcal banks. However, SEEG as described by the authors requires stereotactic stereoscopic teleangiography, a technique that most North American centers do not routinely perform. The “learning curve” in this technology is unclear. In addition, up to 20 depth electrodes were inserted in each patient, a number of brain tissue passes that would make many stereotactic/epilepsy surgeons uncomfortable.

While SEEG is a successful localizing technique, it is not clear that this very invasive monitoring provides any benefit over subdural electrode arrays in combination with the judicious use of specifically targeted depth electrodes.

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This interesting paper covers indications, results, and complications of SEEG recordings in a large number of patients with refractory epilepsy. It describes the extensive experience of a large European epilepsy surgery group using an invasive presurgical evaluation method, which is very popular in the French schools of epilepsy surgery and used much less in the Anglo-American sphere. This paper is of interest because it describes a large experience and includes a number of interesting cases. For example, a high proportion of cases had negative magnetic resonance imaging (28.6%) and a high number of multilobar resections (34%). The authors have evaluated a complicated group of patients and are able to provide outcomes that are subgrouped for different resection techniques. Although the procedures appear relatively invasive, intracerebral hemorrhage with severe permanent deficits only occurred in two cases (1%). An additional 4.2% of patients presented hemorrhagic complications. In summary, this is a complicated patient group, which has been subjected to an extensive, invasive evaluation protocol with a mean of 12 stereotactically implanted depths electrodes. The authors achieved a very respectable seizure freedom rate considering the complicated composition of the patient group.

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